

Target validation paths — demo-nsc lc-egfr-l858r-post-osi-d3m0

The diagnostic and biomarker workup that hardens the targetable-feature call

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Target validation paths

EGFR L858R, MET amplification (GCN 8.2), and T790M-absent are all confirmed at decision-relevant resolution, so no diagnostic test gates the rank-1 trial. The validation work here is about resistance characterization at progression, not target confirmation. The goal is to make sure the case really is MET-amplified bypass resistance and not something more complicated wearing the same imaging signature.

EGFR L858R

The most consequential miss right now would be SCLC histologic transformation, which is the highest-PPV resistance pattern that no liquid biopsy can confirm. Tumor NGS including TP53 and RB1 status is the screen: co-loss is the strongest predictor of transformation, with reported rates of 3–14% across post-osimertinib cohorts. If imaging morphology shifts or co-loss is detected, fresh re-biopsy with a neuroendocrine IHC panel (chromogranin, synaptophysin, INSM1, Ki-67) is the only way to confirm. ctDNA covering EGFR C797S and exon-20 inserts is the parallel on-target resistance check; a positive C797S result reframes the next-line conversation toward fourth-generation EGFR-TKIs or combination strategies.

MET amplification

The FISH GCN 8.2 result clears SAFFRON's enrollment threshold (≥ 6) cleanly, but MET amplification can be focal, and primary-vs-metastatic discordance is a documented confounder in this setting. A second-site FISH (or a matched archival block) refines confidence that the dominant disease, not just the biopsied site, is the amplified one. MET IHC (clone SP44) is the orthogonal-modality complement to the SAFFRON eligibility profile. A comprehensive NGS panel that picks up HER2 / BRAF / FGFR co-alterations frames durability expectations and the next-line plan.

Where to order these assays

Assay	Provider	Contact
ctDNA panel (EGFR C797S / T790M / exon-20)	Guardant Health (Guardant360 CDx)	test info · 1-855-698-8887
ctDNA panel	Foundation Medicine (FoundationOne Liquid CDx)	test info · 1-888-988-3639
ctDNA panel	Tempus (xF+)	test info · 1-800-739-4137
ctDNA panel	Caris Life Sciences (Caris Assure)	test info · 1-888-979-8669
ctDNA panel	NeoGenomics Laboratories (NeoLAB Liquid)	test info · 1-866-776-5907
Tumor NGS for TP53 / RB1 + bypass alterations	Foundation Medicine (FoundationOne CDx)	test info · 1-888-988-3639
Tumor NGS	Caris Life Sciences (Molecular Intelligence)	test info · 1-888-979-8669
Tumor NGS	Tempus (xT)	test info · 1-800-739-4137
Tumor NGS	NeoGenomics Laboratories (NeoTYPE Comprehensive)	test info · 1-866-776-5907

Tumor NGS	Memorial Sloan Kettering (MSK-IMPACT)	test info · 1-212-639-2000
MET FISH (gene/CEP7 ratio)	NeoGenomics Laboratories	test info · 1-866-776-5907
MET FISH	Mayo Clinic Laboratories	test info · 1-800-533-1710
MET FISH	LabCorp / Esoterix Oncology	test info · 1-800-345-4363
MET IHC (clone SP44)	Mayo Clinic Laboratories	test info · 1-800-533-1710
MET IHC (clone SP44)	ARUP Laboratories	test info · 1-800-242-2787
Neuroendocrine IHC panel (re-biopsy)	Mayo Clinic Laboratories	test info · 1-800-533-1710
Neuroendocrine IHC panel	ARUP Laboratories	test info · 1-800-242-2787

Decision support, not medical advice. Confirm assay availability and current standards with the treating team and the local pathology service.